

Infosys Springboard Virtual Internship 6.0

Completion Report

Team Details

Batch Number: 1

Start date: 13th August 2025

Name: **Komal Joshi**

Internship Duration: 8 Weeks

1. Project title

Heart Shield: An AI-Powered System for Early Heart Disease Risk Prediction.

2. Project Objective

The primary objective of the Heart Shield project was to develop a robust and user-Friendly web application that predicts an individual's risk of developing heart disease. The system aims to automate risk assessment by allowing users to either manually input their health data or upload medical documents (like lab reports) for automatic data extraction using Optical Character Recognition (OCR). By leveraging machine learning on key health indicators such as cholesterol, blood pressure, and age, the project provides a clear risk categorization (Low, Moderate, High), making early detection more accessible.

3. Project description in detail

Approach:

Our approach was structured into three core milestones over eight weeks:

1. **Data Engineering & Model Development (Weeks 1-4):** We began with a comprehensive data collection and preprocessing phase, handling missing values and outliers. A key innovation was the development of a custom OCR pipeline to parse data from scanned medical reports. We then engineered features and implemented multiple machine learning algorithms (Random Forest, SVM, Neural Networks and Logistic regression.), systematically optimizing them through hyperparameter tuning and cross-validation to select the best-performing model.
2. **Backend & API Development (Weeks 5-6):** We built a secure Flask/Django backend with a database to manage users and predictions. The core of this phase was integrating the trained ML model into a robust API and implementing a full user authentication system (Sign-up/Login) with role-based access.
3. **Frontend Development & Integration (Weeks 7-8):** We created a responsive and intuitive web interface. The frontend allows for manual data input and document

upload, connects to the backend APIs, and displays the prediction results through an interactive dashboard. The final phase involved end-to-end testing and bug fixes.

Technology Used:

- **Backend:** Python, Flask, SQLite
- **Machine Learning:** Scikit-learn, TensorFlow/Keras, Pandas, NumPy
- **OCR & Document Processing:** Tesseract, OpenCV, Pytesseract
- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Version Control:** Git, GitHub, VS Code IDE

Impact in real world implementation:

- **Improved Access:** Democratizes early heart disease screening, making it accessible to communities with limited healthcare resources.
- **Proactive Prevention:** Enables early identification of at-risk individuals, allowing for timely lifestyle changes and medical intervention before conditions become critical.
- **Cost Reduction:** Lowers long-term healthcare costs by focusing on affordable prevention rather than expensive emergency treatment and chronic disease management.
- **Clinical Support:** Provides doctors with a data-driven tool for risk assessment, supporting and enhancing their clinical judgment.
- **Operational Efficiency:** Automates the extraction of patient data from medical documents, saving time for medical staff and reducing manual entry errors.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Environment Setup, Data Collection & Exploration	Completed setup, downloaded and analysed heart disease datasets (e.g., Cleveland, Framingham).
Week 2	Data Preprocessing, OCR Development, Feature Engineering	Built data cleaning pipeline, developed OCR for medical documents, and created new features.
Week 3	Model Architecture Research, Multi-Algorithm Implementation	Researched and implemented Logistic Regression, Random Forest, SVM, and Neural Network models.
Week 4	Model Evaluation, Hyperparameter Tuning, Risk Categorization Logic	Optimized models, implemented a risk categorization system (Low/Moderate/High), and finalized the best model.
Week 5	Backend Framework Setup, Model Integration, User	Built Flask backend, integrated the ML model as an API, and

	Authentication.	created a secure login system.
Week 6	File Upload System, OCR-Backend Integration, Frontend Core UI	Enabled document uploads, connected OCR to the backend, and built the login and data input forms.
Week 7	Frontend Advanced Features, System Integration	Developed the results dashboard, integrated all components, and began end-to-end testing.
Week 8	Comprehensive Testing, Bug Fixes, Final Documentation & Presentation	Performed final testing, fixed bugs, and prepared the final report and presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Understood project requirements and finalized the technology stack and work distribution.	Week 1
Prototype/First Draft	Completed the first end-to-end pipeline: data input -> ML model -> basic risk prediction in a console.	End of Week 4
Mid-Term Review	Presented a working ML model with >85% accuracy and a functional backend with core APIs.	End of Week 4
Final Submission	Completed a fully integrated and tested web application with all features.	End of Week 8
Presentation	Prepared and delivered the final project demonstration and submitted all documentation.	End of Week 8

5b. Project execution details

The project was executed systematically through three distinct phases, following an iterative development approach.

Phase 1: Data Foundation & Model Development

- **Data Engineering:** Established a robust data pipeline involving collection, comprehensive cleaning (handling missing values, outliers), and exploratory analysis of cardiovascular datasets. A key component was developing an OCR module using Tesseract and OpenCV to automatically extract medical metrics from scanned reports.

- **Feature Engineering & Training:** Processed features through scaling and encoding to make them model-ready. We then implemented and evaluated multiple algorithms—including Logistic Regression (baseline), Random Forest, SVM, and Neural network—using stratified cross-validation.
- **Model Optimization & Selection:** Systematically tuned hyperparameters and employed ensemble methods (Voting, Stacking). The best-performing model was selected based on accuracy, precision-recall, and ROC-AUC metrics, achieving over 85% accuracy. A clinical risk categorization system (Low/Moderate/High) was implemented using calibrated probability thresholds.

Phase 2: Backend & Core System Development

- **Server & API Setup:** Configured a Flask backend with a structured database (User, Prediction, Medical Record) and built secure REST API endpoints.
- **Model & Auth Integration:** Seamlessly integrated the serialized ML model and preprocessing pipeline into the backend to ensure consistent predictions. Implemented a full user authentication system with JWT tokens, role-based access (Patient/Doctor), and password encryption.
- **Document Handling:** Enabled secure medical file uploads and integrated the OCR pipeline for asynchronous text extraction and data parsing, auto-filling the input form for user review.

Phase 3: Frontend Development & End-to-End Integration

- **Responsive UI/UX:** Developed an intuitive, mobile-friendly frontend using Bootstrap/Tailwind. Key pages included a dashboard for manual input and file upload, a results page with clear risk visualization (color-coded badges, confidence scores), and a user history log.
- **System Integration & Testing:** Connected all frontend components to backend services. Conducted thorough end-to-end testing of the complete user journey—from registration and login to prediction and history viewing—resolving bugs to ensure a reliable and polished final product.

6. Snapshots / Screenshots

Register Page:



The screenshot displays the 'Heart Shield AI' registration interface. The header features the brand name 'Heart Shield AI' on the left and navigation links 'Back to Dashboard', 'Login', and 'Register' on the right. The main content area is split into two sections. On the left, there is a large, stylized illustration of a red heart being held by a hand, with a red line representing a heartbeat or medical line. On the right, there is a 'Register' form with the following fields: 'Username' (with a hint 'Choose a unique username'), 'Full Name' (with a hint 'Enter your full name'), 'Email address' (with a hint 'Enter your email'), 'Password' (with a hint 'Create a strong password'), and 'Confirm Password' (with a hint 'Confirm your password'). A red 'Register' button is positioned below the form fields. At the bottom of the form, there is a link: 'Already have an account? [Login here](#)'.

Login Page:

Heart Shield AI

[Back to Dashboard](#) [Login](#) [Register](#)

Welcome to Heart Shield

Login

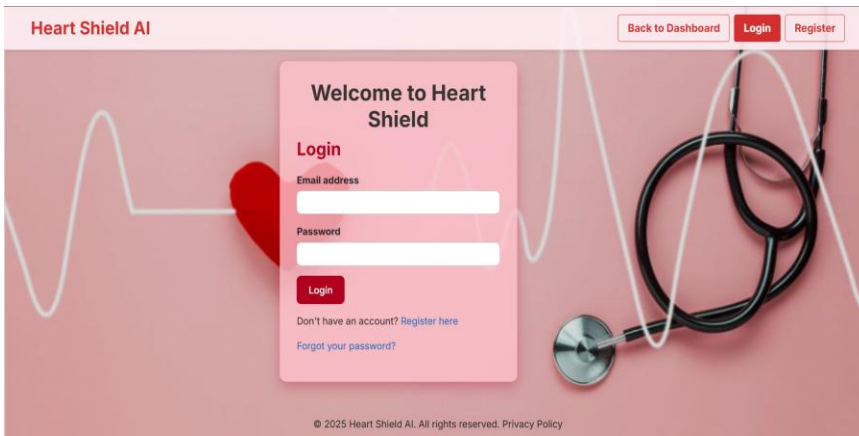
Email address

Password

Login

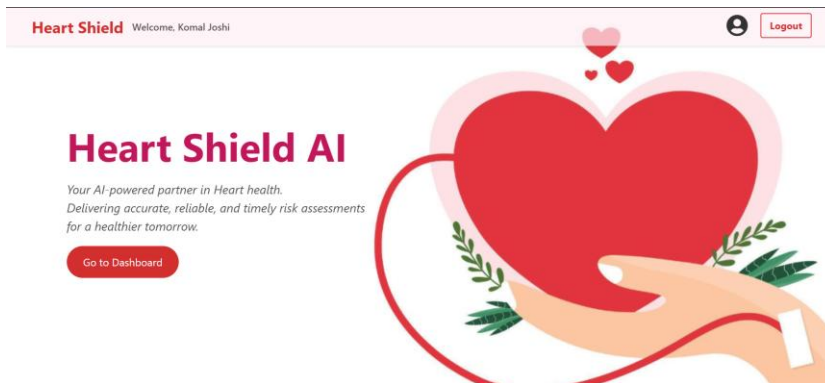
Don't have an account? [Register here](#)

[Forgot your password?](#)

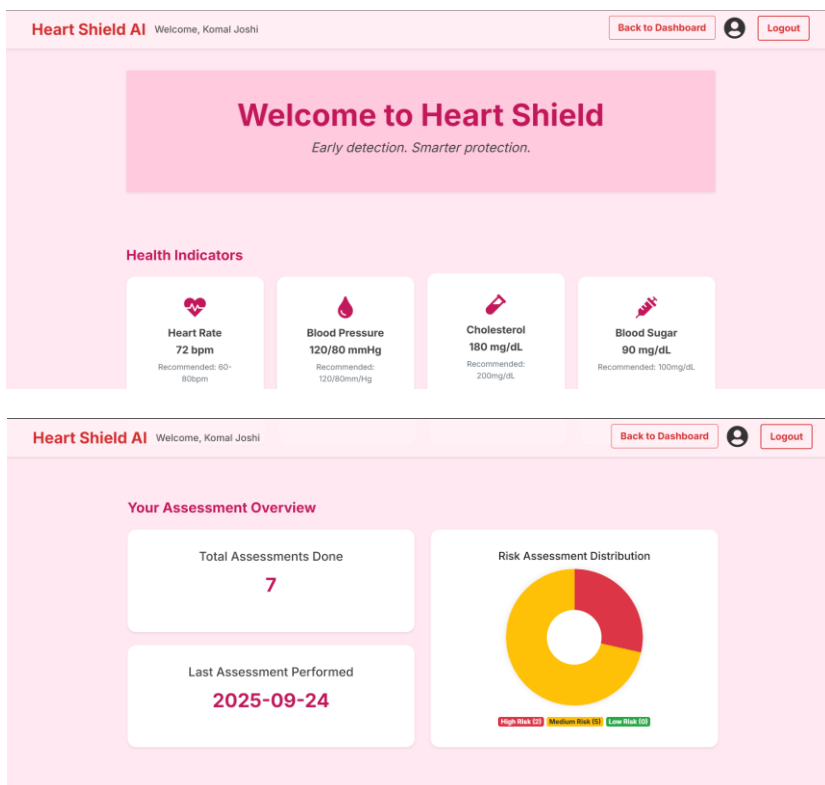


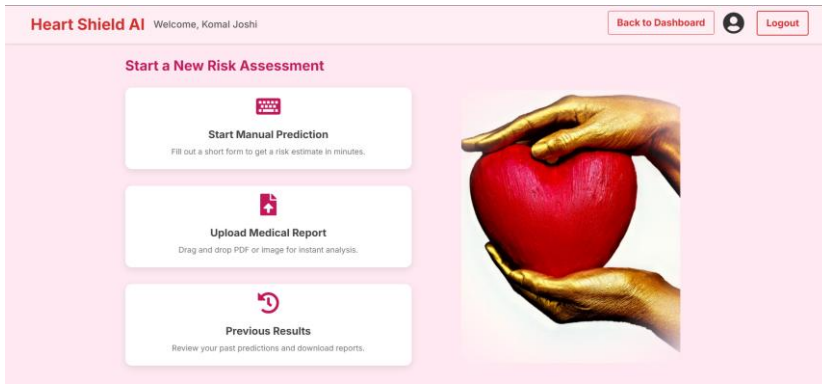
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Home Page:

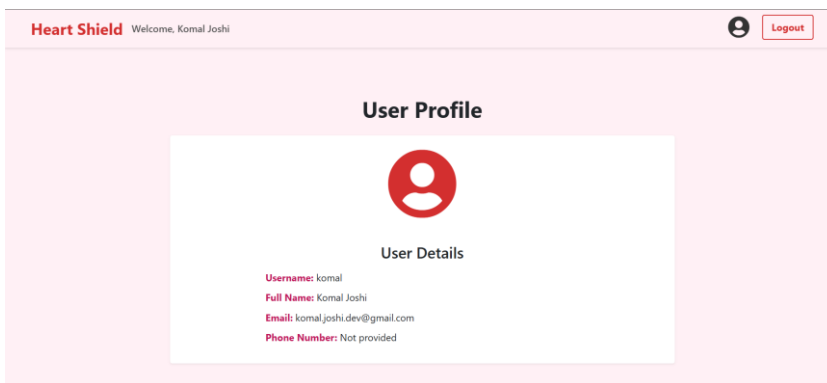


Dashboard:

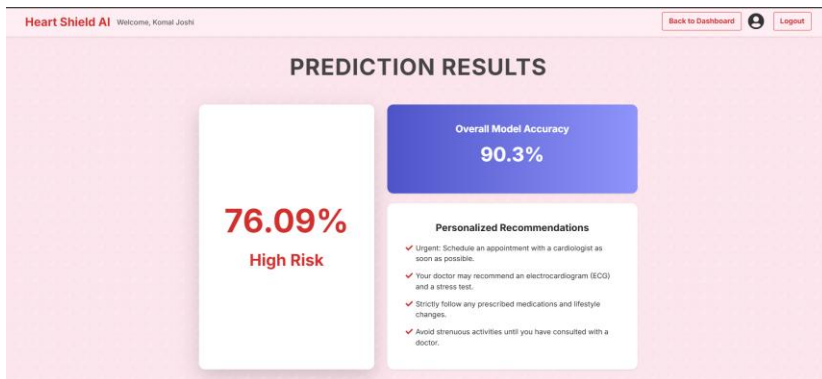




User profile:



Result Page:



Extracted OCR output

```
(venv) PS C:\Users\komi\Desktop\coding\Komal-Joshi> & C:/Users/komi/Desktop/coding/Komal-Joshi/venv/Scripts/python.exe c:/Users/komi/Desktop/coding/Komal-Joshi/src/ocr_utils.py

Extracted Text

MEDICAL REPORT
Age : 28
Sex : Female
Chest pain : Non anginal pain
Blood pressure : 135
Cholesterol : 230
Blood sugar : 0
ECG : Abnormal
Max heart rate : 145
Exercise angina : No
ST depression : 1.2
ST slope : Flat
Thalassemia : Reversible defect

ocr data

{'Age': '28', 'Sex': 'Female', 'chestPain': 'Non anginal pain', 'bloodPressure': '135', 'Cholesterol': '230', 'bloodSugar': '0', 'ECG': 'Abnormal', 'maxHeartRate': '145', 'exerciseAngina': 'No', 'stDepression': '1.2', 'stSlope': 'Flat', 'Thal': 'Reversible defect'}

Numeric Data

{'Age': 28, 'Sex': 0, 'chestPain': 3, 'bloodPressure': 135, 'cholesterol': 230, 'bloodSugar': 0, 'ECG': 1, 'maxHeartRate': 145, 'exerciseAngina': 0, 'stDepression': 1.2, 'stSlope': 2, 'Thal': 3}
(venv) PS C:\Users\komi\Desktop\coding\Komal-Joshi>
```

7. Challenges Faced

- Challenge 1 (Technical - OCR Accuracy):** Initially, the OCR had low accuracy in reading numeric values from poorly scanned documents.

Resolution: We implemented image preprocessing techniques (noise reduction, contrast enhancement) and created post-processing validation rules to check for physiologically plausible values (e.g., realistic blood pressure ranges).
- Challenge 2 (Technical - Data Imbalance):** The dataset had slightly imbalanced classes, which could lead to biased predictions.

Resolution: We used stratified splitting during train-test splits and focused on metrics like Precision, Recall, and F1-score instead of just accuracy. We also experimented with techniques like SMOTE during training.
- Challenge 3 (Operational - Integration):** Integrating the separate ML, backend, and frontend components led to unexpected errors and data format mismatches.

Resolution: We held daily stand-up meetings to sync progress and used a "contract-first" approach for APIs, where the frontend and backend teams agreed on the data JSON structure before development began.

8. Learnings & Skills Acquired

- **Technical Skills:**

- **Machine Learning:** Hands-on experience with the full ML lifecycle, from data preprocessing and feature engineering to model training, evaluation, and deployment.
- **Web Development:** Proficiency in full-stack development using Flask/Django for backend and HTML/CSS/JS for frontend.
- **OCR Technology:** Gained expertise in using Tesseract and OpenCV to solve a real-world data extraction problem.
- **Tools:** Git for version control, Pandas for data manipulation, Scikit-learn for ML models.

- **Soft Skills:**

- **Project Management:** Learned to break down a complex project into manageable tasks and adhere to a strict timeline.
- **Collaboration & Communication:** Effectively worked in a team, dividing tasks and integrating individual components into a cohesive final product.
- **Problem-Solving:** Enhanced ability to debug complex issues spanning multiple system layers.

9. Testimonials:

Working on Heart Shield was an incredible journey. Taking a project from a concept to a fully functional application that can potentially impact lives was immensely rewarding. I particularly enjoyed the challenge of integrating the ML model into a live web system and seeing it provide real-time predictions.

10. Conclusion:

The Heart Shield project was a comprehensive and successful internship experience that aligned perfectly with our academic studies in computer science and data analytics. It allowed us to apply theoretical knowledge to a practical, impactful real-world problem in the healthcare domain. We are proud to have delivered a functional application that demonstrates the power of AI in preventive healthcare. This experience has solidified our career goals in AI and full-stack development and provided us with a strong portfolio piece.

11. Acknowledgements:

We would like to express our sincere gratitude to **Infosys Springboard** for providing us with the invaluable opportunity to work on the "AI-Powered Application for Early Detection of Heart Disease Risk" project.

A special note of thanks to our mentor, **Gunadeep sir**, for his expert guidance, unwavering support, and insightful feedback throughout the 8-week internship. His mentorship was

instrumental in navigating the challenges of the AI/ML technology stack and successfully completing our milestones.

We are also thankful to our program coordinator, **Disha ma'am**, for seamlessly managing the schedule, communication, and resources, ensuring a smooth and organized internship experience.

Finally, we extend our appreciation to our entire team for their dedication, collaboration, and hard work, which made the successful completion of this project a rewarding journey.